HOSPITALITY REVENUE ANALYSIS

Python Project

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FULL PROJECT OVERVIEW

INTRODUCTION

- Atliq Grands is a prominent hotel chain in India, operating across major cities such as Delhi, Mumbai, Bangalore, and Hyderabad. With over 20 years of experience in the hospitality industry, it offers multiple hotel categories (Atliq Seasons, Atliq Exotica, Atliq Bay, and Atliq Palace) along with diverse room types such as Standard, Elite, Premium, and Presidential.

- Despite its strong market presence, the company has recently experienced a decline in revenue and market share. To address this challenge, Atliq Grands is leveraging data analytics to enable data-driven decision-making and uncover actionable insights aimed at driving revenue growth and improving overall competitiveness.

PROBLEM STATEMENT

- Atliq Grands is experiencing a decline in revenue and market share due to increasing competition in the hospitality industry.

- This project aims to analyze booking data from both direct and third-party platforms to identify key trends and generate actionable insights that can improve revenue and overall business performance.

REVENUE GAP & COMPETITION

DATA OVERVIEW

We have 5 csv files

1) dim_date.csv :
- It contains date, mmm yy, week no, day_type

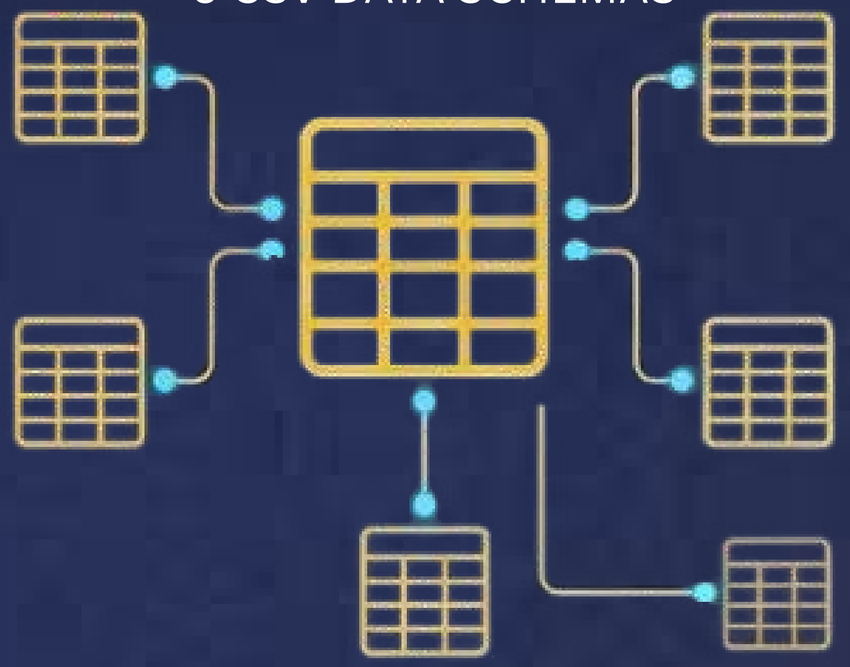
2) dim_hotels.csv:
- It contains property_id, property_name, category, city

3) dim_rooms.csv:
- It contains room_id, room_class

4) fact_aggregated_bookings.csv:
- It contains property_id, check_in_date, room_category, successful_bookings, capacity

5) fact_bookings.csv:
- It contains booking_id, property_id, booking_date, check_in_date, checkout_date, no_guests, room_category, booking_platform, ratings_given, booking_status, revenue_generated, revenue_realized

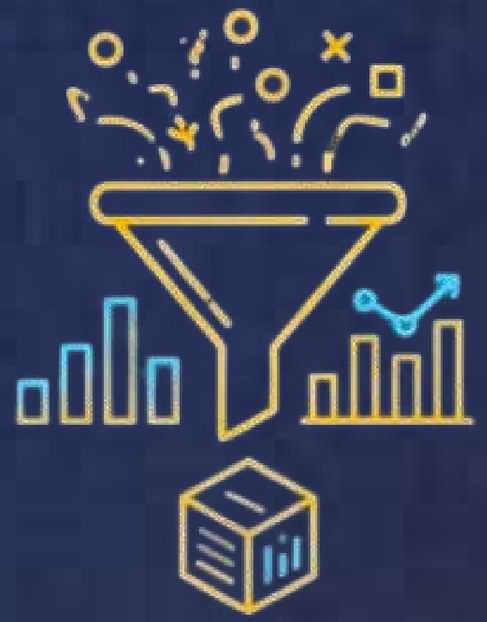
5 CSV DATA SCHEMAS



DATA PREPARATION & TRANSFORMATION

- Clean Invalid Data
- Outlier Detection and Removal
- Creating New Columns

PANDAS PIPELINE



KEY INSIGHTS

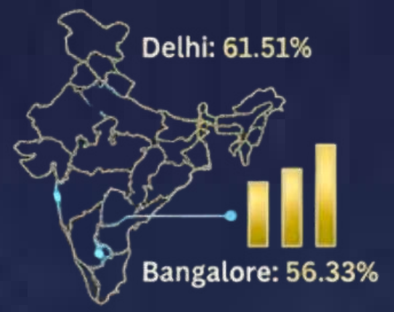
1. WHAT IS THE AVERAGE OCCUPANCY RATE IN EACH OF THE ROOM CATEGORIES?

- Among the room categories, Presidential rooms have the highest average occupancy rate (59.28%), indicating slightly higher demand compared to other categories, while Standard rooms show the lowest occupancy.

```
df.groupby('room_class')['occ_pct'].mean().round(2)
```

room_class	occupancy %
Elite	58.01
Premium	58.03
Presidential	59.28
Standard	57.89

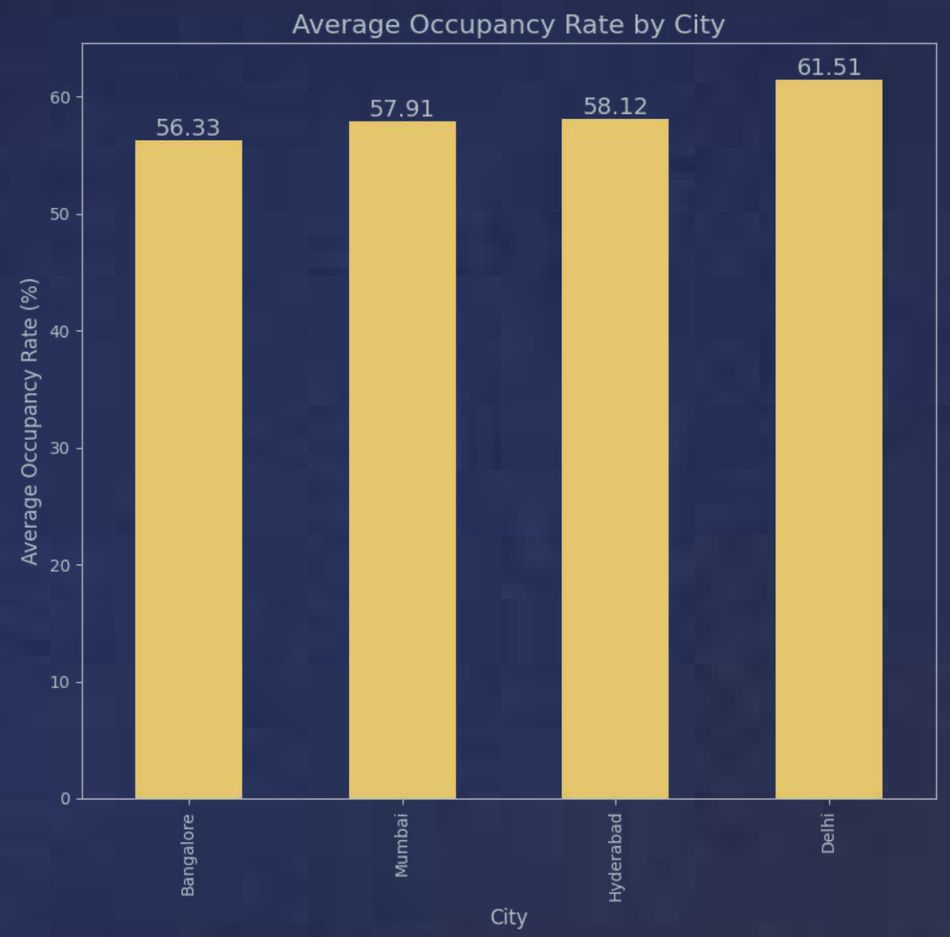
KEY INSIGHTS



2. WHAT IS THE AVERAGE OCCUPANCY RATE PER CITY

- Delhi has the highest average occupancy rate among cities (61.51%), while Bangalore has the lowest (56.33%)

```python
graph = df.groupby('city')['occ_pct'].mean().sort_values()

num_bars = len(graph)

fig_width = max(8, num_bars * 0.5)
fig_height = 8

plt.figure(figsize=(fig_width, fig_height))

ax = graph.plot(kind = 'bar', facecolor = '#e6c76e')

plt.xlabel('City',color='#B0BEC5', fontsize=12)
plt.ylabel('Average Occupancy Rate (%)',color='#B0BEC5', fontsize=12)
plt.title('Average Occupancy Rate by City',color='#B0BEC5', fontsize=16)

ax.tick_params(axis='x', colors='#B0BEC5')
ax.tick_params(axis='y', colors='#B0BEC5')

for spine in ax.spines.values():
    spine.set_color('#B0BEC5')

for index, value in enumerate(graph):
    plt.text(index, value, str(round(value, 2)), ha = 'center', va = 'bottom', fontsize = 14,
color='#B0BEC5')

plt.tight_layout()

plt.savefig('average occupancy rate per city.png', transparent = True, bbox_inches='tight')

plt.show()
```



KEY INSIGHTS

3. WHEN WAS THE OCCUPANCY BETTER? WEEKDAY OR WEEKEND?

- Occupancy rates are significantly higher on weekends compared to weekdays, highlighting an opportunity to implement targeted marketing strategies and dynamic pricing to maximize revenue

```
df.groupby('day_type')['occ_pct'].mean().round(2)
```

day_type	occupancy %
weekday	50.88
weekend	72.34

KEY INSIGHTS

4. IN THE MONTH OF JUNE, WHAT IS THE OCCUPANCY FOR DIFFERENT CITIES

- In June, Delhi recorded the highest occupancy rate (62.47%), followed by Hyderabad (58.46%), indicating strong demand in these markets.

```python
graph = df_june.groupby('city')['occ_pct'].mean().round(2).sort_values(ascending = False)

num_bars = len(graph)

fig_width = max(8, num_bars * 0.5)
fig_height = 8

plt.figure(figsize=(fig_width, fig_height))

ax = graph.plot(kind = 'bar', facecolor = '#e6c76e')

plt.xlabel('City',color='#B0BEC5', fontsize=12)
plt.ylabel('Average Occupancy Rate (%)',color='#B0BEC5', fontsize=12)
plt.title('Average Occupancy Rate in June by City',color='#B0BEC5', fontsize=16)

ax.tick_params(axis='x', colors='#B0BEC5')
ax.tick_params(axis='y', colors='#B0BEC5')

for spine in ax.spines.values():
    spine.set_color('#B0BEC5')

for index, value in enumerate(graph):
    plt.text(index, value, str(round(value, 2)), ha = 'center', va = 'bottom', fontsize = 14,
color='#B0BEC5')

plt.tight_layout()

plt.savefig('average occupancy rate in june by city.png', transparent = True, bbox_inches = 'tight')

plt.show()
```

KEY INSIGHTS

5. WHAT IS THE REVENUE REALIZED PER CITY

- Mumbai generates the highest revenue among all cities

```
df_bookings_all.groupby('city')['revenue_realized'].sum()
```

city	revenue_realized
Banglore	420383550
Delhi	294404488
Hyderabad	325179310
Mumbai	668569251

KEY INSIGHTS

6. WHAT IS THE MONTH BY MONTH REVENUE

- Revenue trends indicate monthly fluctuations, with a peak in May 2022 (₹40,83,75,641) and a decline observed in July 2022 (₹38,99,40,912).

```
df_bookings_all.groupby('mmm yy')['revenue_realized'].sum()
```

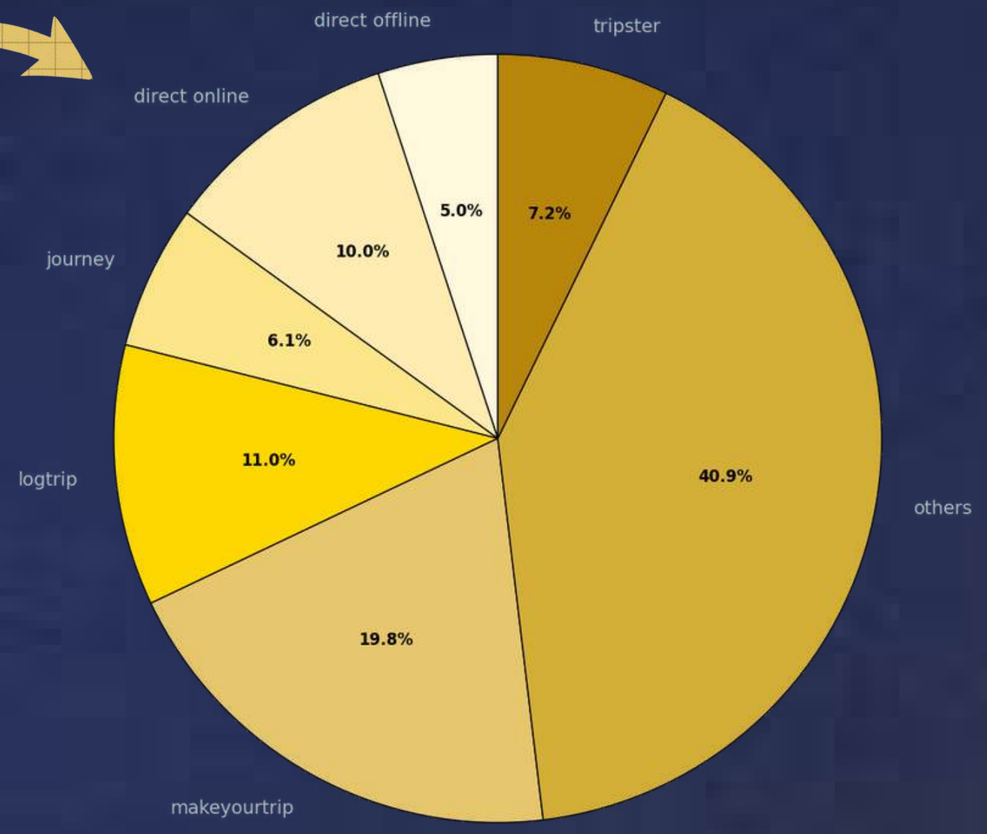
mmm yy	revenue_realized
Jul-22	389940912
Jun-22	377191229
May-22	408375641

KEY INSIGHTS

7. PRINT A PIE CHART OF REVENUE REALIZED PER BOOKING PLATFORM

- The revenue realized per booking platform varies, with others being the highest contributor , followed by makeyourtrip.

```python
graph = df_bookings_all.groupby('booking_platform')['revenue_realized'].sum()

num_categories = len(graph)
fig_width = 8 + (num_categories * 0.5)
fig_height = 8 + (num_categories * 0.5)

plt.figure(figsize=(fig_width, fig_height))

wedges, texts, autotexts = plt.pie(
    graph, labels=graph.index, autopct='%1.1f%%', startangle=90, colors = ['#FFF8DC', '#FFECB3',
'#FDE68A','#FFD700', '#E6C76E', '#D4AF37', '#B8860B' ]   * len(graph),
    wedgeprops={
        'edgecolor': 'black', 'linewidth': 1})

for text in texts:
    text.set_fontsize(14)
    text.set_color('#B0BEC5')

for autotext in autotexts:
    autotext.set_fontsize(12)
    autotext.set_color('black')
    autotext.set_weight('bold')

ax.set_ylabel('')

plt.figtext(0.5, 0.05, 'Revenue Realized per Booking Platform', ha = 'center', va = 'center', fontsize
= 16,color='#B0BEC5')

plt.axis('equal')

plt.savefig('revenue realized per booking platform.png', transparent = True, bbox_inches = 'tight')

plt.show()
```



Revenue Realized per Booking Platform

RECOMMENDATIONS

- Increase Direct Bookings : Promote bookings through the official website to reduce third-party commission costs and improve overall profit margins.
- Implement Dynamic Pricing Strategy : Adjust room pricing based on demand patterns, seasonality, and occupancy trends to maximize revenue.
- Weekend-Specific Offers : Design exclusive weekend packages, deals, and experiences to attract more short-term and leisure travelers.
- Expand in High-Performing Markets : Invest in expanding presence in high-demand cities like Delhi by opening new properties or enhancing existing ones to increase market share and revenue.

THANK YOU

FOR WATCHING MY DATA DRIVEN STRATEGY PRESENTAION